

Biometric ATM Authentication using Machine Learning

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Abstract — Biometric ATMs are now widely utilised by many people. However, carrying your ATM card around with you is really tough. Users may find themselves unable to access their funds if their ATM card becomes destroyed. There aren't any that are suitable. Authentication mechanisms that are used to ensure the security of ATM transactions. This document has been improved with basic biometric identification technologies and focuses on ATM security procedures. The goal is to create a model for an effective automatic identification system that uses facial recognition and OTP to simplify the management of financial transactions.

Keywords—ATM, Biometric Authentication, OTP, Tkinter.

I. INTRODUCTION

As science and technology advance, future creations will be developed with a high level of security. On the contrary, Threats, on the other hand, are made to breach this level of protection. Overall, though, the advancement of technology has had a favourable impact. Various financial organisations, such as banks, and apps, such as ATMs, are nevertheless vulnerable to fraud and theft. Fraudsters are seeking new ways to strike as technology advances, improving security. Biometric identifiers have a number of advantages over traditional and current authentication and security methods. PIN-based verification is used exclusively in the newer ATM security authentication mechanism. Emergency situations, pin reminders, interaction speed, and unintentional pin sharing all have an impact on the system. Magnetized cards Cloning chips is simple. The opposite sides of the same coin are security and the unsecured; an automated machine becomes vulnerable owing to a security flaw. ATM makers bolster and enhance security measures so that customers may conduct banking transactions with ease and without worry of funds being syphoned from their accounts. This project's major goal is to improve the security of the standard ATM module. We've developed a new assumption approach that enhances the overall ATM transaction experience, usefulness, and convenience. To improve user account security and privacy, features like face recognition and one-time passwords (OTP) are implemented. Facial recognition technology allows the machine to recognise each user individually making the master key the face As a result, the possibilities are completely ruined as a result of ATM card theft and copying. The LRR algorithm was used in the establishment of the ATM system for the face. While verifying the true user of the account, the user is recognised.

The method for authentication that uses:

- Image processing: The open cv will capture images of user using webcam it will process the image of the user.

II. LITERATURE REVIEW

Praveena P , Et al [1] propose the The Face recognition method's heart, principal component analysis (PCA), finds a linear combination of features that maximises the overall variance in the dataset. It's a powerful way of encoding data, but it doesn't consider every class, so when components are removed, a lot of identifying information is lost.

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Darwin Nesakumar A1, Et al[3] developed a s can be done using a USB camera and a fingerprint module. If analysis matched, the ATM will allow the transaction. Only after Entering this OTP into this system allows the user to withdraw funds otherwise procedure will be end. Now let's talk about the processing performed in this security system.

Mala K [4] developed the integration of biometrics into the banking application has the ATM payment gateway is now more dependable and secure. The addition of the OTP idea to the system improves security and eliminates the need for us to remember passwords. Furthermore, the system is based on embedded technology, making it both user-friendly and non-intrusive. The ATM terminal is protected from theft thanks to this mechanism.

Pratiksha Shetiya, Et al [5]. The ability to recognise and compare a user's face (IRIS) of eyes is being developed. The database took pictures and sent them to the computer system, where they were processed using MATLAB routines. The output iris image was then compared to the database iris image, and if they were the same, the person had access to the system or something else. The user request is turned down.

III. METHODOLOGY

The purpose of this paper is to find a Machine Learning model to identify the real time face recognition. In this system we use pycharm which simplifies setup and access as well as UI design for using the tkinter framework in Python.

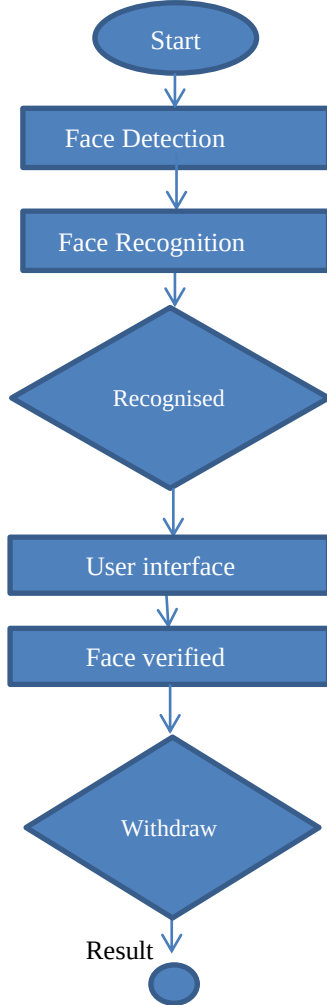


Figure 1: Biometric ATM Authentication The steps that require to be followed are:

A. Requirements

- Import packages.
- Pycharm-software interface.
- Tkinter
- Webcam

B. Import Opencv&Dlib: I've used OpenCV and dlib a lot for face detection and recognition and dlib is much more accurate than the OpenCV based face detector Haar.

C. OTP verification: This is the step that identifies the user and we set an email id and password and the system verifies the user for security purposes.

D. Machine Learning: Here machine to learn the user faces. The face is identified by frames at real-time from the data array that had trained before.

IV. IMPLEMENTATION

Model creation is the main step in face recognition and OTP verification. First step is open a Pycharm tkinter project then, Import the packages that are necessary.

```
from pathlib import Path
from collections import Counter
# import the necessary packages
from sklearn.preprocessing import LabelEncoder
from sklearn.svm import SVC
from imutils.video import VideoStream
from imutils.video import FPS
import time
from tkinter import *
from tkinter import messagebox
import sqlite3
import pandas as pd
from PIL import Image, ImageTk
import tkinter as tk
import pandas as pd
```

Figure 2: Import the packages that are necessary.

```
self.frame.destroy()
self.frame = Frame(self.root, bg="black", width=900, height=500)
root.geometry("900x500")
self.enroll = Button(self.frame, text="Enroll", bg="#50A880", fg="white", font=ARIAL, command=self.enroll_user)
self.withdraw = Button(self.frame, text="Withdraw Money", bg="#50A880", fg="white", font=ARIAL, command=self.withdraw_money)
self.q = Button(self.frame, text="Quit", bg="#50A880", fg="white", font=ARIAL, command=self.root.destroy)
self.enroll.place(x=0, y=315, width=200, height=50)
self.withdraw.place(x=600, y=315, width=200, height=50)
self.q.place(x=340, y=340, width=120, height=20)
self.frame.pack()
```

Figure 3: Load Frame for picture and learn and recognize it.

```
def withdraw_money_page(self):
    self.frame.destroy()
    self.frame = Frame(self.root, bg="black", width=900, height=500)
    # Login Page Form Components
    self.label1 = Label(self.frame, text="Note:", bg="black", fg="white", font=ARIAL)
    self.label2 = Label(self.frame, text="1. By clicking on the 'Verify Face Id' button, we proceed to perform facial recognition.", bg="black", fg="white", font=ARIAL)
    self.label3 = Label(self.frame, text="2. Each capture will take 15 seconds are you are required to move your face in.", bg="black", fg="white", font=ARIAL)
    self.label4 = Label(self.frame, text="3. If your face is recognized, you will be required to input your account pass.", bg="black", fg="white", font=ARIAL)
    self.label5 = Label(self.frame, text="4. If your face is not recognized after 5 seconds, you will automatically be logged out.", bg="black", fg="white", font=ARIAL)
    self.label6 = Label(self.frame, text="5. If your face is not recognized after three trials, you won't be allowed to withdraw money.", bg="black", fg="white", font=ARIAL)
    self.label7 = Label(self.frame, text="6. To begin, click the 'Verify Face Id' button below", bg="black", fg="white", font=ARIAL)
    self.button = Button(self.frame, text="Verify Face Id", bg="#50A880", fg="white", font=ARIAL, command=self.verify_face_id)
    self.q = Button(self.frame, text="Quit", bg="#50A880", fg="white", font=ARIAL, command=self.root.destroy)
    self.b = Button(self.frame, text="Back", bg="#50A880", fg="white", font=ARIAL, command=self.begin_page)
    self.frame.pack()
```

Figure 4: Create Payout Window .

```
self.frame.pack()
data = pd.read_csv('bank_details.csv')
```

Figure 5: Read csv file to fetch bank details .

V. RESULT

The result shows that a specific user's face is recognized. Face recognition has proven to be one of the most secure methods of all biometric systems to a point of high security and preventing ATM robberies and providing ATM security. It will first authorize the use of the user face, if it matches then it will be pushed to the UI and OTP sent and verified. It will use the help of tkinter framework in python.



Figure 6: ATM user interface.

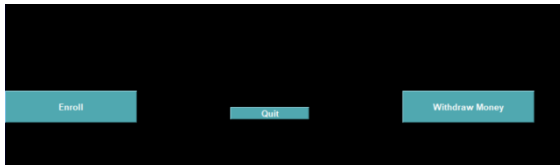


Figure 7: Choose options.

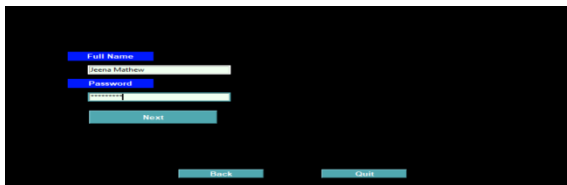


Figure 8: User Enrollment



Figure 9: Message Box.

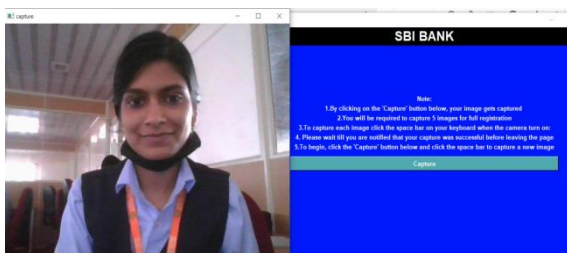


Figure 10: Face recognition for user identification.

of illegal transactions at ATM points without the knowledge from the genuine owner. Using a biometric for identification is strong.

VII. REFERENCES

- [1] Praveena.P, Savithri.V; Saratha.R, Monisha.M and Ashwini.R. Face Recognition Open CV based ATM security system. International Journal for Modern Trends in Science and Technology 2021.
- [2] Pooja Surawse, Sonam Bhange, Shreya Taru, Samruddhi Khot, Secure ATM Transactions Using Face Recognition & OTP Department of Computer Engineering, Indira College of Engineering And Management Parandwadi, Pune, India, JETIR (Journal of Emerging Technologies and Innovative Research 2022 February.
- [3] Darwin Nesakumar A1, T Suresh, Nivedha T, K Nivedha S, Priyadharshini G, P Mugilan. Smart ATM Security Using Face Recognition R.M.K. Engineering College, Department of ECE12345, European Journal of Molecular & Clinical MedicineSSN 2515-8260 Volume 7, Issue 4, 2020
- [4] Mala K ,ATM SECURITY BY USING BIO-METRIC METHOD,Computer Science and Engineering, SSIT, Tumakuru, Karnataka.www.ijcrt.org © 2018 IJCRT | Volume 6, Issue 1 January 2018 | ISSN: 2320-2882
- [5] Pratiksha Shetiya,Meryl Mascarenhas,Mrunal Deshmukh ATM Security System using Iris Recognition by Image Processing International Journal of Engineering Research & Technology (IJERT),July-2020

VI. CONCLUSION

This paper aims to use biometrics to make the ATM transaction structure increasingly reliable and secure. The idea added to the framework further improves the security. I developed an ATM model that is more authentic by ensuring security using facial recognition software. In this project, we tried to propose a solution to the problem worry about the problem of fraudulent Transactions through the ATM through biometric data, which are only possible if the account holder is physically there. Thus, it cancels the cases